

Your grade: 100%

Next item →

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1.

If $|\Sigma| = 2$, how many elements are in Σ^6 ?

1 / 1 point

☐

∅

☐

36

☐

12

☒

64

✔ Nice work

Correct. It is all strings with length 6, so $2*2*2*2*2*2$ is 64, which is the number of strings with length 6.
2.

The state of a finite automaton can be represented graphically with a:

1 / 1 point

☒

Node

☐

Transition

☐

Function

☐

Arc

✔ Nice work

Correct. States are represented by nodes.
3.

Finite automata use which of the following to describe transitions?

1 / 1 point

☐

Nodes

☐

Concatenation

☐

Sigma

☒

Arcs

✔ Nice work

Correct. We use arcs to represent transitions.
4.

What is the difference between a deterministic and non-deterministic finite automaton?

1 / 1 point

☒

The next possible state is clearly set in a deterministic finite automaton. A non-deterministic finite automaton may have many next possible states.

☐

Deterministic finite automaton can transition to and be in multiple states at once for a given input.

☐

Non-deterministic finite automaton can transition to but cannot be in multiple states at once for a given input.

☐

They are identical.

✔ Nice work

Correct.
5.

Consider an NFA designed to accept strings with an odd number of 'a's and an even number of 'b's. Which of the following statements correctly describe the behavior of this NFA with respect to the given language criteria?

1 / 1 point

☐

The NFA has a loop that returns to its start state with every 'a' read, ensuring an odd count of 'a's is accepted.

☒

The NFA transitions between two states with 'b's, ensuring an even number of 'b's.

☐

The NFA accepts strings with an odd number of 'a's only when followed directly by a 'b'.

☐

The NFA accepts any string with 'b's due to its non-deterministic nature.

☐

The NFA accepts an odd number of 'a's by transitioning to a dead state, rejecting strings with even 'a's.

✔ Nice work

Think about how the NFA handles 'b's and what it means for them to be even.